

University of Eloued
Academic Year: 2025/2026

Lesson: Linear Regression and Logistic Regression

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Short Summary and Series:01

Linear Regression (Simple and Multiple Linear regression)

Introduction to Linear Regression

Linear regression is a fundamental supervised learning method used to model and quantify the relationship between an independent variable X and a continuous dependent variable Y . In simple linear regression, the relationship is represented by the equation :

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where β_0 is the intercept, β_1 is the slope coefficient, and ϵ represents the error term. The main objective is to estimate the coefficients and use the resulting model to make predictions. **The main rules are:** the dependent variable should be continuous; the observations should be appropriately paired; the independent and dependent variables must have the same number of observations; an intercept should normally be included in the model; and the estimated relationship should be evaluated using appropriate measures such as R^2 , residuals, and prediction errors. In this section, these principles are demonstrated through a simple practical example.

Formula

If you mean the formulas involving the mean values $\bar{X}$ and $\bar{Y}$, they are:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$$

And the coefficient formulas can also be written as:

$$\hat{\beta}_1 = \frac{\sum X_i Y_i - n \bar{X} \bar{Y}}{\sum X_i^2 - n \bar{X}^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

1. Determination of the Model ?

Calculating model performance and goodness-of-fit for a linear regression model involves evaluating error magnitudes and variance explained using three standard metrics.

Mean Squared Error (MSE)

MSE measures the average squared difference between the actual observed values (y_i) and the predicted values $\hat{y}_i$ generated by the model.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ✓ Subtract each predicted value ($\hat{y}_i$) from its corresponding actual value (y_i) to find the residual for each observation.
- ✓ Square each residual to eliminate negative signs and heavily penalize larger errors.
- ✓ Sum all the squared residuals.
- ✓ Divide the total sum by the number of observations (n)

2. Root Mean Squared Error (RMSE)

RMSE is the square root of the Mean Squared Error, returning the error metric back to the original units of the dependent variable for intuitive interpretation.

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

3. Coefficient of Determination (R^2)

R^2 represents the proportion of variance in the dependent variable that is predictably explained by the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

(Where $\bar{y}$ is the mean of the observed values, the numerator is the Residual Sum of Squares [RSS], and the denominator is the Total Sum of Squares [TSS].)

4. Practical Example

1. Festival try to do Simple Linear regression using just one Independent variable .
2. and in the second try using python apply Multiple linear régression

1. First Example: Linear regression ,Concept and application

Dataset for Linear Regression

Depended variable: Exam_Score

Obs	Hours_Studied	Exam_Score
1	1	12
2	2	11
3	3	13
4	4	10
5	5	16
6	6	17
7	7	14
8	8	18

2. Simple Linear Regression — Questions (use Manually method)

1. What are the independent and dependent variables?
2. Calculate the mean of (X) and (Y).
3. Calculate the regression coefficient (β_1).
4. Calculate the intercept (β_0).
5. Write the estimated regression equation.
6. Predict Exam_Score for a given number of study hours.
7. Calculate the prediction errors (e_i).
8. Calculate the MSE.

9. Calculate the RMSE.
10. Calculate (R^2).
11. Interpret the value of (β_1 and β_0).
12. Interpret the value of (R^2).
13. Plot the observations and the estimated regression line.
14. Estimate the model in Python and compare the results with the manual calculations.

Second Step: Multiple Linear Regression using Short dataset

1. Use coding python to resolve the same example adding Two others variables

Obs	Hours_Studied	Exam_Score	Attendance_Percent	Previous_Score
1	1	12	70	60
2	2	11	75	65
3	3	13	80	68
4	4	10	65	62
5	5	16	85	75
6	6	17	90	80
7	7	14	82	72
8	8	18	95	85

Questions

Answer at the same precedent questions

Third Step: Multiple Linear Regression using large dataset

Dataset: you can find the dataset in Git_Hub link:

https://github.com/atman500/Linear_regression_deep_learning-2026-2027/tree/main

E	D	C	B	A	
Sleep_Hours	Previous_Score	Assignments_Completed	Attendance_Percent	Study_Hours	1
8.2	76.2	10	67.3	2.6	2
7.6	79.4	10	78.4	3.7	3
8.3	100	10	81.3	9.1	4
7.5	87.3	7	66.2	4.3	5
6.9	74.9	10	62.6	4	6
8.4	78.1	10	70.5	4.2	7
6.9	68.2	8	75.4	4.9	8
7.6	97.8	7	72	4	9
6.4	83.8	9	71.6	2.9	10
7.8	85	7	68.2	5.6	11
5.9	76.4	9	75.4	4.6	12
8.1	97.4	9	60.9	4.5	13
5.8	76.7	10	81.9	3.6	14
7.8	100	10	91.1	6.3	15
6.3	94.7	9	81.7	5.5	16
7.2	90.4	10	62	5.5	17
7.8	85.2	10	78.2	4.7	18
7.4	67.6	8	59.6	2.2	19

Questions

Using the same dataset, with `Exam_Score` as the dependent variable and the following independent variables:

- `Study_Hours`
- `Attendance_Percent`
- `Assignments_Completed`
- `Previous_Score`
- `Sleep_Hours`

1. What are the dependent and independent variables?
2. Write the multiple linear regression model.
3. Estimate the Linear model:
4. Write the estimated regression equation.
5. Interpret each regression coefficient.
6. Predict `Exam_Score` for a given student profile.
7. Calculate the prediction errors (e_i).
8. Calculate the MSE.
9. Calculate the RMSE.
10. Calculate (R^2).
11. Calculate Adjusted (R^2).
12. Interpret (R^2) and Adjusted (R^2).
13. Estimate the model in Python using `statsmodels`.
14. Compare the Python results with the calculated results.
15. Which independent variables have the strongest effect on `Exam_Score`?